HONGTAO XIE

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RESEARCH INTERESTS

3D computer vision and neural rendering (Gaussian Splatting, NeRF); generative models for human motion and human–object interaction; contrastive representation learning; efficient sequence architectures (RetNet, state-space models) for vision and multimodal learning.

EDUCATION

Brown University Sep 2026 – May 2028 (expected)
Sc.M. in Electrical and Computer Engineering *Providence, RI*

Beijing Normal–Hong Kong Baptist University (BNBU) Sep 2022 – Jun 2026
B.S. (Honors) in Computer Science and Technology *Zhuhai, China*
Undergraduate thesis: *Efficient Text-Driven 3D Human–Object Interaction Synthesis with Retention-Based Diffusion Models*. Advisor: Prof. Amy Hui Zhang.

PUBLICATIONS

[C1] **H. Xie** and H. Zhang. "SCGC: Sparse Control-Guided Gaussian Splatting with Contrastive Learning." In *Proceedings of the 20th ACM SIGGRAPH International Conference on Virtual-Reality Continuum and its Applications in Industry (VRCAI)*, 2025. DOI: 10.1145/3779232.3779453

[W1] **H. Xie** and S. Zhu. "MambaGuard-MOO: Multi-Objective Safe Drug Recommendation with Contrastive Learning." In *IEEE BigData 2025 Workshop on AI in Big Data Analytics for Healthcare*, 2025.

RESEARCH EXPERIENCE

RetHOI: Text-Driven 3D Human–Object Interaction Synthesis [Code] Jun 2025 – Dec 2025
Undergraduate Thesis Researcher · Advisor: Prof. Amy Hui Zhang *BNBU*
Guangdong Provincial Key Laboratory of Interdisciplinary Research and Application for Data Science

- Investigated whether retention-based sequence modeling can jointly address three weaknesses of diffusion-based HOI generators: long-range temporal dependencies, physically plausible contact, and slow sampling.
- Replaced the dual Transformer branches of HOI-Diff with parallel RetNet branches coupled by bidirectional cross-retention between human and object trajectories; added multi-scale Gabor filtering for contact-trajectory refinement and CLIP-guided contrastive alignment in both the affordance-prediction and motion-generation stages.
- On BEHAVE (1,451 interaction sequences), reduced foot-skating ratio by **94%** and contact distance by **21.3%** relative to HOI-Diff, improved FID from 1.501 to 1.443, and cut inference time by **31.1%** (27.3 → 18.8 min per 196-frame sequence on an RTX 4090). Unpublished.

SCGC: Sparse Control-Guided Gaussian Splatting [Code] Sep 2024 – Dec 2025
Independent Researcher · Advisor: Prof. Amy Hui Zhang *BNBU*

- Introduced explicit structural guidance into Scaffold-GS through 512 sparse control points, initialized by farthest point sampling and linked to scene anchors via KNN, with periodic control-point updates and anchor reassignment during training.
- Designed a two-layer, four-head Transformer that refines control-point geometry from global spatial context, and a dual-level contrastive objective over control points and Gaussian primitives to improve multi-view consistency.
- Across eight NeRF-Synthetic scenes, the first-author paper [C1] reports mean **33.61 dB PSNR / 0.951 SSIM** (+0.07 dB / +0.019 over Scaffold-GS), with the clearest gains in regions with complex geometry or little texture. Later re-implemented from scratch and re-evaluated over multiple seeds on four benchmarks; code released.

MambaGuard-MOO: Multi-Objective Safe Drug Recommendation [Code] May 2025 – Aug 2025
Independent Researcher · Advisor: Prof. Jude Shengxin Zhu *BNBU*

- Extended the MambaDrug state-space model to encode longitudinal diagnosis and procedure histories, injecting drug–drug interaction (DDI) constraints and medication–diagnosis associations into patient-state transitions.

- Formulated an adaptive multi-objective loss balancing efficacy, safety, ranking, and personalization, combined with drug-embedding contrastive learning (safe positives vs. therapeutically similar, higher-risk negatives) introduced through staged training.
- Against nine baselines on MIMIC-III, achieved **F1 0.7052** (vs. 0.6418 for SafeDrug) while **halving the DDI rate** (0.0682 → 0.0341). First-author paper [W1].

Monocular 4D Dynamic Scene Reconstruction Jun 2024 – Dec 2024
Remote Research Project · Mentor: Heng Yu, Ph.D. Student, Stanford University *Remote*

- Reviewed NeRF and 3D Gaussian Splatting literature on extending static scene representations to dynamic scenes captured by a single moving camera.
- Prepared and standardized datasets, corrected camera poses, trained a model to initialize point clouds from 2D images, and explored time-varying Gaussian offsets and rotations for dynamic reconstruction.

Quantifying Semantic Understanding in Machine-to-Machine Communication Mar 2024 – May 2024
Research Assistant · Advisor: Prof. Ricky Yuen-Tan Hou *BNBU*

- Designed and validated a hierarchical metric of semantic understanding for M2M communication, analyzing its behavior across sentence lengths and polysemy conditions.
- Proposed a WordNet-based word-sense probability formulation that improved semantic consistency on complex and ambiguous sentences; contributed to a multi-level feedback mechanism for verifying transmitted information.

Lightweight 2D Human Pose Estimation Sep 2023 – Dec 2023
Research Assistant · Advisor: Prof. Shuhong Chen *Guangzhou University*

- Profiled Lite Pose and related lightweight architectures (parameters, FLOPs, module-level cost), used sensitivity analysis to guide pruning and quantization, and benchmarked runtime, memory, and energy on edge devices (Raspberry Pi, mobile NPUs).

Selected Projects

Sign Language Video-to-Text Recognition Oct 2024 – Dec 2024

- Built a 3D-CNN recognizer for continuous sign-language video (ResNet baseline) and used BERT-based semantic completion to turn fragmented recognition output into coherent sentences.

Chest X-ray Diagnostic Text Generation Oct 2024 – Dec 2024

- Trained a ResNet classifier for chest X-ray findings and used BERT to convert predicted labels into structured diagnostic text.

Honors and Awards

President's Honor Roll, BNBU — Spring 2025, Fall 2025
Dean's List, BNBU — Fall 2022, Fall 2023, Spring 2024
Special Mentions Prize (HKD 3,000), HKBU Greenhack Challenge "Hack for Climate Action" — 2023
Global Leader Programme in AI and Computer Science, HKU Institute for China Business — 2023

Technical Skills

Languages: Python, C++, C, CUDA
Frameworks: PyTorch, OpenCV, Open3D, NumPy, scikit-learn
Methods: diffusion models, 3D Gaussian Splatting, NeRF, Transformers, RetNet, Mamba / state-space models, CLIP, contrastive learning
Tools: Linux, Git, GitLab